Sky Maps and Catalogues: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

10 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IOAA questions or marking schemes. Parameters are supplied locally; an exam's constants take precedence.

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Solution 1 — RA to degrees. Return →

6 h is $6 \times 15 = 90^\circ$. Conversely, 90° is $90/15 = 6$ h. **Check:** 24 h corresponds to 360°.

Solution 2 — Catalogue fields. Return →

Check the identifier, coordinate epoch/equinox, photometric band and magnitude, positional uncertainty, and proper motion if the time baseline is long. **Check:** a position without an epoch is incomplete.

Solution 3 — Proper motion. Return →

The motion is $0.5 \times 50 = 25''$. A $1'$ field is $60''$ across, so $25''$ is about 42% of the full field width—large enough to matter for precise pointing or narrow-field imaging. **Check:** high-proper-motion stars can move much more.

Solution 4 — Chart scale. Return →

The separation is 3.5°. A mirrored chart reverses east and west, so position angles and star-hopping directions must be flipped. **Check:** many telescope finders or diagonal optics produce mirrored images.

Solution 5 — Magnitude limit. Return →

The ratio is $10^{0.6(8-6)} = 10^{1.2} = 15.8$. **Check:** going two magnitudes deeper multiplies the number of stars by about 16 in this simple model.

Solution 6 — Epoch. Return →

Precession changes the coordinate grid over decades, and proper motion changes the star's position relative to the grid. Parallax and aberration matter for high precision. **Check:** J2000 is a reference frame, not a permanent pointing direction.

Solution 7 — RA separation. Return →

For stars at the same declination and a small RA difference, two minutes of RA is $2 \times 15' = 30'$ along the equator. At declination 60°, the east–west separation is approximately $30' \cos 60^\circ = 15'$. **Check:** the cos δ factor accounts for converging RA lines.

Solution 8 — Cross-identification. Return →

The entries could be a physical binary, a chance alignment, a variable star observed at different phases, or a catalogue error. Compare epochs, bands, proper motions and images; a common proper motion or resolved image can decide. **Check:** never assume the brighter entry is the target.

Solution 9 — Finder-chart design. Return →

Use a field of about 15–30′ if the telescope field is small, with a magnitude limit near 11–12 so the target and comparisons are visible but the field is not overwhelmed. Mark north and east, state the epoch, and choose several non-variable comparison stars of similar colour and magnitude. Include a wider inset chart for star-hopping. **Check:** the chart must match the actual eyepiece or camera orientation.

Solution 10 — Selection effect. Return →

A magnitude-limited sample includes intrinsically bright stars from large distances and faint stars only nearby, so it does not sample a fixed volume. A volume-limited sample requires distances and a cutoff within which all stars of the relevant type are included. **Check:** Malmquist bias is the classic consequence of magnitude-limited sampling.